

3- B Tech

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Faculty of Technology & Engineering
University Backlog Examination May- 2011
EC-203 Solid State Electronics

Date: 19/05/2011

Time: 10.00 a.m. To 1.00p.m

Maximum Marks: 70

Instructions:

1. Attempt all the Questions.
2. Figure to the right indicates full marks.
3. Make suitable assumptions and neat figures if required.

SECTION-I

Q-1(a) Discuss the following terminologies in brief. [10]

- (1) Mass Action Law
- (2) Diffusion current
- (3) Reverse Saturation current
- (4) Field effect transistor
- (5) Phototransistor

Q-2 (a) For JFET, if $I_{DSS} = 20 \text{ mA}$, $V_p = -5 \text{ V}$ and $g_{mo} = 4 \text{ mA/V}$. Determine the transconductance for $V_{GS} = -4\text{V}$. [03]

Q-3(a) Explain Continuity Equation. [08]

(b) Explain FET Static characteristics. [06]

OR

Q-3(a) Explain Hall Effect. [08]

(b) Explain Energy band theory of crystals. [06]

Q-4 Attempt any two questions. [08]

- (1) PN junction as a rectifier
- (2) Classification of Amplifiers
- (3) Full wave Rectifier

[P.T.O.]

SECTION- II

- Q-5 Attempt any four questions.
- (1) Explain fixed biasing of transistor.
 - (2) Compare FET over BJT.
 - (3) Thermal runaway
 - (4) Typical characteristics of CE input Characteristics
 - (5) Avalanche Multiplication

- Q-6 (a) Derive all h-parameters from hybrid small signal model for common emitter configuration. [07]
- (b) For the transistor shown in figure A. The value of β is 100. Calculate the value of R_B . $V_{CE(sat)} = 0.3$ V. [08]

OR

- Q-6 (a) Draw the load line for the figure B. What is I_c at saturation point? Find V_{CE} at cut-off point. [08]
- (b) Using sketches for voltage divider biased CB amplifier, mention guidelines for converting CB transistor amplifier to its h-parameters model. [07]
- Q-7 Attempt any two.
- (1) Explain emitter follower.
 - (2) Transistor Current components.
 - (3) Thermistor compensation.
 - (4) Explain Self bias. [12]

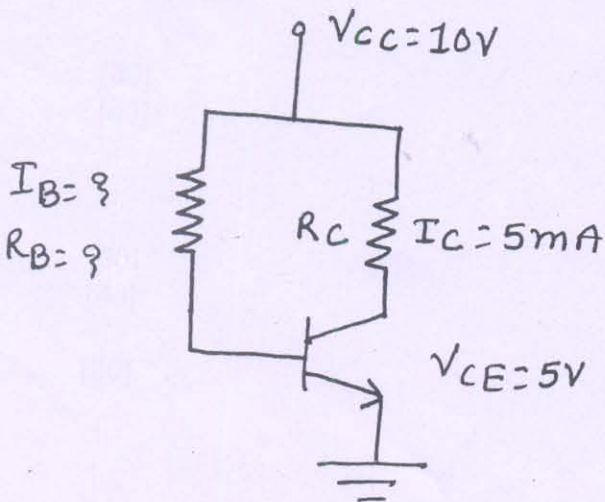


Figure A

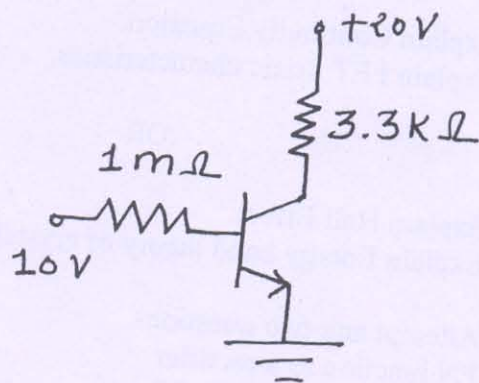


Figure B